CRISTIAN BUDALA

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Computer Science student at the University of Bucharest, with a prior degree in Business Administration and Economics. Interested in Computational Research, Graph Theory and Extremal combinatorics and Causal & Graph methods for AI Governance.

EDUCATION

University of Bucharest, *BSc in Mathematics and Computer Science | Bucharest, Romania* *2025 - 2028*

Awards: 1st place winner at the 2026 Scientific Communications Session, Certificate of volunteer work at the Institute for Artificial Intelligence, Romanian Academy.

***Academy of Economic Sciences Bucharest**, BA in Business Administration and Economics | Bucharest, Romania* *2021 - 2024*

Awards: 1st place winner of The Web Campaign competition, Award of completion of social entrepreneurship incubator at University of Tallinn, Speaker at the 2024 Scientific Communications Session

EXPERIENCE

L'Oreal Romania, *Insight & Analytics Intern CDMO Team* *Feb 2023 - Oct 2023*

- Initiated Project RCPE in Romania, implementing multilingual NLP capabilities including sentiment analysis, profanity filtering, and intent classification for consumer-engagement queries across social media channels.
- Owned consumer-engagement analytics and insights; performed exploratory data analysis, process mining, predictive analytics, and regression modeling for e-commerce projects.
- Contributed to the L'Oreal and Google Privacy Sandbox project for first-party advertising data and collaborated with Google, Meta, TikTok, and research partners.

PUBLICATIONS

- "Sunflower-Free Uniform Families: Recursive Constructions and Explicit Bounds". E. Axante, C. Budala, D. Chitic, B.C. Dumitru, and M. Nacu. (2026). 10.48550/arxiv.2609.06175.
- "An Exhaustive Census of Main-Diagonal Symmetric Costas Arrays of Orders 37-42". B.C. Dumitru, C. Budala. (2026). 10.48550/arXiv.2608.28690.
- "Local Patterns and the Cost of Costas Array Search". B.C. Dumitru, C. Budala. (2026). hal-05731061.

PROJECTS

Autonomous Multi-Model Discovery & Causal AI Governance Platform (Erdős #1038 & #36) | *Python, FLINT/Arb, SciPy/CVXPY, OpenAI & Gemini APIs*

- Architected a closed-loop multi-agent research platform (up to 64 concurrent subagents, 695 sessions, 1.88B tokens) that resolved the 68-year-old Erdős Problem #1038 and improved the world-record lower bound margin on Erdős Problem #36 by 5x (0.38056870).
- Implemented causal reasoning governance: SHA-256 prompt fingerprinting, full thought-chain logging across 1,147 multi-model jobs (61.5% Gemini Pro/Flash, 38.5% GPT-6 Astra/Sol), cross-model peer review, and 25 hostile adversarial audits to eliminate LLM sycophancy.

Costas Array Enumerator | *C++, CUDA, Slurm, OpenMP*

- Designed and implemented a backtracking solver for enumerating Costas arrays of order 30, using symmetry reduction, forward checking, and prefix decomposition.
- Optimized the search for large-scale HPC execution and developed a CUDA split-and-conquer implementation for GPU acceleration.

8-bit Movfuscator | *C and x86 Assembly*

- Co-developed an educational compiler that transforms arithmetic, logical, and control-flow operations into programs driven almost entirely by x86 MOV instructions and lookup tables, proving Turing-completeness.

Multilingual Sentiment Analysis and Intent Classification | *Python, RoBERTa, DistilBERT*

- Built a Romanian-to-English sentiment pipeline reaching 90% accuracy and trained a 13-class intent model on 6,500 synthetic and augmented queries, reaching 73% test accuracy.

TECHNICAL SKILLS

Computation: C/C++, CMake, x86 Assembly, RISC-V, CUDA, Git, Linux, Bash, Zsh, Lua, LaTeX, OpenMP, Slurm | **AI/ML:** Python, PyTorch, Keras, Transformers, BERT, R, Tensorflow, scikit-learn, LangGraph, FastAPI